

Homework (Carolina Reaper)

- Q1.** A music producer mixes 2 parts of a fast rhythm pattern with 3 parts of a slow rhythm pattern. What is the ratio of fast to total rhythm patterns?
(A) 2 : 5
(B) 3 : 5
(C) 2 : 3
(D) 1 : 2
(E) 3 : 2
- Q2.** A painter mixes a 15% red paint solution with a 30% red paint solution to obtain 12 ounces of a 21% red paint solution. How many ounces of the 15% solution are used?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q3.** A DJ mixes 150 songs with a rhythm frequency of 120 beats per minute (bpm) with 50 songs with a rhythm frequency of 100 bpm. What is the average rhythm frequency of the mixed songs?
(A) 112 bpm
(B) 114 bpm
(C) 116 bpm
(D) 118 bpm
(E) 120 bpm
- Q4.** A musician records a 3-minute song with a 1 : 2 ratio of fast to slow notes. If the song has 180 notes, how many fast notes are there?
(A) 45
(B) 60
(C) 72
(D) 90
- (E) 120
- Q5.** An artist paints a 4 : 5 ratio of red to blue pixels on a 20×30 canvas. How many blue pixels are there?
(A) 120
(B) 150
(C) 180
(D) 240
(E) 300
- Q6.** A composer writes a 5-minute melody with a 2 : 3 ratio of high to low notes. If the melody has 240 notes, how many low notes are there?
(A) 96
(B) 120
(C) 144
(D) 160
(E) 192
- Q7.** A band mixes a 25% instrumental song with a 50% instrumental song to obtain a 30% instrumental song. What is the ratio of the first song to the second song?
(A) 1 : 2
(B) 1 : 3
(C) 2 : 3
(D) 3 : 4
(E) 4 : 5
- Q8.** A painter mixes 2 gallons of a 30% blue paint solution with 1 gallon of a 50% blue paint solution. What percentage of the mixture is blue?
(A) 32%
(B) 34%
(C) 36%
(D) 38%
(E) 40%

Open Ended Questions (FRQ)

- Q9.** A music producer mixes two types of sound effects, A and B, with the following characteristics: A has a 20% chance of being a 100-decibel sound and an 80% chance of being a 50-decibel sound. B has a 30% chance of being a 120-decibel sound and a 70% chance of being a 40-decibel sound. If the producer mixes 3 parts of sound effect A with 2 parts of sound effect B, what is the probability that the resulting sound effect will be a 100-decibel sound?
- Q10.** An artist paints a 12×18 canvas with a 3 : 5 ratio of blue to red pixels. If the artist then adds a 2×3 section with a 2 : 3 ratio of blue to red pixels, what is the new ratio of blue to total pixels on the canvas?

Chef's Tips

- Tip 1: Encourage students to use equations to solve mixture problems
- Tip 2: Remind students to interpret concentration and ratios in the context of arts and music
- Tip 3: Emphasize the importance of concise and unique problem-solving